KACNet: Kolmogorov–Arnold Convolution Network for Hyperspectral Anomaly Detection

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***Abstract*— Hyperspectral images capture numerous narrow spectral bands to provide detailed information to identify and locate targets, making them highly suitable for anomaly detection tasks. In recent years, deep learning techniques have demonstrated impressive capabilities and prospects in hyperspectral anomaly detection (HAD), primarily relying on multilayer perceptrons (MLPs) and convolutional neural networks (CNNs) to extract and represent nonlinear features. However, MLPs and CNNs often require deeper network architectures when dealing with complex high-dimensional data, resulting in a constrained generalization and limited representation of features. To address this issue, and inspired by the recent Kolmogorov–Arnold network (KAN), this article introduces a novel asymmetric convolutional autoencoder (AE) network by integrating KAN and CNN, named `KACNet`. Specifically, we design a spectral KAN block in the convolutional encoder and a spatial KAN block in the convolutional decoder, to simultaneously enhance the feature extraction and characterization capabilities of the network. Furthermore, to effectively utilize the limited prior information, a weight initialization mechanism based on hierarchical density-based spatial clustering of applications with noise (HDBSCAN) is developed to boost the background recovery. By combining KAN, CNN, and HDBSCAN, the proposed integration enhances the interpretability and reliability of HAD. Extensive experiments are conducted on six public datasets, demonstrating that the KAN poses remarkable performance on background reconstruction, particularly, the proposed `KACNet` significantly outperforms the other state-of-the-art methods.**

***Index Terms*— Anomaly detection, convolutional autoencoder (AE), deep learning, hyperspectral images (HSIs), Kolmogorov–Arnold network (KAN).**

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I. Introduction

HYPERSPECTRAL images (HSIs) contain not only 2-D spatial location information per pixel but also rich spectral information across the electromagnetic spectrum, covering a wide range of wavelengths from the visible to the near-infrared [1], [2], [3]. Since different materials exhibit different spectral responses at different wavelengths, HSIs with detailed spectral information are capable of distinguishing objects with similar appearances but different material compositions [4], [5], [6], which is particularly important for the detection of anomalous substances [7]. In many real-world applications where the prior information about the target object is not known, unsupervised hyperspectral anomaly detection (HAD) methods are remarkable and compelling techniques to locate and detect potentially anomalous targets, and they have been extensively applied in various fields, such as environmental monitoring, agricultural surveillance, national defense and security, and medical diagnostics [8], [9].

Notwithstanding the capabilities of HAD techniques, the challenges posed by HSIs are notable, as they provide abundant information for image analysis which, at the same time, places a major hurdle due to the processing of high-dimensional data with hundreds of channels [10]. In HAD tasks, anomalies are generally characterized by a low probability of occurrence, small size, and significant deviation from the surrounding background spectrum. The background typically dominates the image, and the distribution depends on different application scenarios with a level of complexity and variability. In this regard, how to extract useful information from high-dimensional HSIs for background and anomaly separation is a persistent challenge [11]. Up to now, the developed HAD methods can be briefly categorized into traditional and deep learning methods, where the former mainly includes statistical theories and linear reconstruction techniques for HAD, and the latter mainly includes popular and effective networks such as convolutional neural networks (CNNs), autoencoders (AEs), generation adversarial networks (GANs), and Transformers.

A. Traditional HAD Approaches: Statistical-Based and Representation Learning-Based Techniques

According to the characteristics of background (wide distribution) and anomalies (sparse and different from the

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surrounding background) in HSIs, utilizing statistical methods is the most intuitive and simple way to detect anomalies. One of the classical pioneers is the RX algorithm proposed by Reed and Yu [12], which assumes the background satisfies a Gaussian distribution. Anomalies are determined by constructing a covariance matrix for the background modeling and calculating the Mahalanobis distance of each pixel from the background model. The larger the Mahalanobis distance, the greater the difference between the pixel under test (PUT) and the background, i.e., it is more likely to be an anomaly. Despite the simplicity and generality of RX, the Gaussian distribution assumption does not exactly match the complex nonuniform background in real applications. To this end, local RX is designed to handle background nonuniformity by modeling the background in a local window [13]. The subspace RX detector decreases the computational complexity of the covariance matrix by projecting the data into a low-dimensional subspace [14]. Kernel RX introduces a kernel function to capture the nonlinear relationship in HSIs for the improvement of detection performance [15]. More recently, novel statistical models have also been applied in HAD tasks, such as the Gaussian mixture model [16], kernel isolation forest [17], two-step generalized likelihood ratio test [18], and so on, which enhance the detection accuracy and robustness.

With the continuous development of signal processing and machine-learning techniques, HAD algorithms based on representation learning have become another typical representative of traditional methods, especially sparse representation (SR), collaborative representation (CR), and low-rank representation (LRR). The main idea for this type of HAD method is to reconstruct the background via a representation model and a matching dictionary, and then take the representation residuals as the anomaly degree of the PUT. Specifically, the SR-based HAD method designs an overcomplete background dictionary to reconstruct the background under a set of sparse bases [19], [20]. In contrast, the CR-based HAD method utilizes the collaboration between the dictionary atoms rather than the competition to reconstruct the background, which is simpler to compute and more suitable for HAD tasks. Given this, Li and Du propose a CR detector (CRD) that models the background by conducting CR in a local dual-window and designed a kernel version of CRD to exploit the nonlinear feature [21]. Vafadar and Ghassemian then propose a modified version of CRD by purifying the background dictionary to enhance the stability of the detection result [22]. Besides the dictionary contamination problem, some recent studies have been devoted to solving the model constraint problem by performing CRD in each feature dimension [23], [24], which mines feature differences to improve detection accuracy.

However, both SR and CR rely on local information to rebuild the background, ignoring the effect of global information. Considering highly correlated structures of the background, the LRR-based method separates anomalous pixels by constructing background low-rank models in which the background part is constrained by low-rank and the anomaly part is constrained by sparsity. In terms of 2-D data processing, Sun et al. [25] decompose the 2-D HSI into three components including background, anomaly, and noise to mitigate the impact of noise while solving the anomaly matrix. Xu et al. [26] build a global background dictionary to recover the background and simultaneously introduced a sparse constraint to the representation coefficient combined with a low-rank constraint, which enhances the recovery of low-rank structure. In terms of 3-D data processing, tensor decomposition is adopted to keep the inherent structural properties of HSI data. Li et al. [27] first exploit higher order robust principle component analysis (RPCA) to separate the HSI into a background and an anomaly tensor. After that, Wang et al. [28] design a 3-D LRR model that reconstructs the background tensor from the view of multiple subspaces. Li et al. [29] introduce a low-rank prior and a piecewise-smooth prior into the background tensor and a spatial group sparse prior into the anomaly tensor, which improves the generality of the method. Despite the under-utilization of features, and compared to the statistical-based methods, this type of method is commonly capable of attaining better detection accuracy due to the simplicity and rationality of the mathematical models.

B. Deep Learning for HAD: Powerful and Promising Solutions

In recent years, deep learning has created a research wave in various fields, as well as new opportunities and challenges for Earth observation research. HSIs have significant application advantages in anomaly detection due to their rich spectral information, but they also possess many issues such as high-dimensional data processing, complex and variable backgrounds, and target diversity. In this case, deep learning can provide a boost to the HAD task through automatic feature extraction, multiscale feature fusion, and nonlinear relationship modeling [11]. Early deep learning networks did not produce satisfactory performance in HAD, as these methods were initially designed primarily for supervised tasks [30]. Subsequently, with the development of self-supervised/unsupervised network architectures, deep learning began to show a broad promising application in HAD. Currently, popular deep learning architectures exploited for HAD tasks are mainly AEs, GANs, and Transformers.

The AEs-based detector reconstructs the HSI by feature squeezing and feature expansion, where anomalous pixels will have larger representation residuals than the background. According to this, Zhao et al. [31] utilize multiple symmetrical encoders and decoders to reconstruct HSIs as well as remove the noise. Lu et al. [32] impose a manifold constraint on the implicit features of the AE by using a local linear embedding method to preserve the intrinsic structure of HSIs. Similarly, graph regularization [33], spectral constraint [34], [35], hyperbolic space and the corresponding wrapped normal distribution [36], and variational AE (VAE) [37] are introduced into AEs to better separate the background and the anomaly. Moreover, convolutional AEs are also developed, which can couple local information to recover the background while compressing features. Wang et al. [38] trained an end-to-end fully convolutional net to reconstruct the background from a random noise map. Gao et al. [39] transferred the low-vision spot network into the HAD task, which can model the

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background while suppressing the representation of anomaly. Furthermore, Wang et al. [40], [41] improved the network to make it more suitable for the detection of multiple types of targets.

The GANs-based detector is an adversarial network consisting of a generator and a discriminator, where the generator learns the distribution of background and generates data resembling the background, while the discriminator attempts to differentiate between the generated and real data. In this process, anomalous pixels tend to have larger reconstruction errors as they cannot be effectively reconstructed by the generator. To model the background distribution of HAD, Xie et al. [42] adopt an unsupervised clustering method to select background samples for the generator to train the network and implement anomaly detection combined with RX. On this basis, Jiang et al. [43] devise a spatial feature extraction module driven by morphological attribute profiles to reinforce the integrity of the anomaly shape; Jiang et al. [44] add discriminators to both hidden and reconstructed features to ensure differential reconstruction of backgrounds and anomalies, realizing an unsupervised GAN framework for HAD. Arisoy et al. [45] integrate the AE and GAN into the HAD framework, and explore its performance in spectral, spatial, and spatial–spectral domains. Wang et al. [46] introduce multivariate Gaussian distribution to inference background generation by combining VAE and GAN with two discriminators.

Different from the aforementioned methods, the transformer-based network captures long-range dependencies through a self-attention mechanism, enabling deeper exploitation of global information in the spectral and spatial dimensions of HSIs. Xiao et al. [47] apply a transformer block with a spatial–spectral dual-window mask to build a HAD network, which can establish associations between pixels or spectral bands on a global scale. Young et al. [48] employ transformers to compute self-attention scores on the abundance maps after low-vision-source separation to guide the generation of anomaly maps. Lian et al. [49] design a gated control strategy for the transformer to achieve the differential reconstruction of backgrounds and anomalies. Wu and Wang [50] exploit an AE framework composed of transformers and the patch-generating method based on clustering to boost the background recovery. Finally, He et al. [51] simultaneously construct a global transformer and a local transformer to realize comprehensive information integration.

C. Motivations and Proposed Contributions

In terms of deep learning-based HAD methods, how to synchronize the reconstruction of the background and the suppression of anomalies is an endless challenge. Existing approaches explore the implementation of HAD in mainstream frameworks such as CNNs, AEs, GANs, and Transformers, which excavate deep features for background reconstruction by introducing attention mechanisms, deepening network structures, and constructing multiscale modules. Meanwhile, regularization constraints, background guidance, low-vision spot mechanism, gate control, and others are developed to inhibit the reconstruction of anomalous targets. Nevertheless, in the face of complex application scenarios and high-dimensional HSIs, the performance of multilayer perceptrons (MLPs) and CNNs in nonlinear feature mining and characterization remains limited and obscured, resulting in unstable and unreliable results. To this end, KAN, a neural network architecture based on the Kolmogorov–Arnold classical theorem, has recently been developed, which assumes that any multidimensional continuous function can be represented as a combination of a series of 1-D continuous functions, thus providing a decomposed representation for high-dimensional inputs. Specifically, KAN applies learnable activation functions to the weights, thus enabling the network to process and learn complex relationships in the input data with a more flexible and closer to the Kolmogorov–Arnold representation theorem. Compared to MLPs and CNNs, KAN has a more powerful nonlinear feature learning capability and physical interpretability, which is crucial for image reconstruction.

Inspired by KAN, in this article, we explore the performance of KAN in HAD tasks and develop a Kolmogorov–Arnold convolutional network (`KACNet`), which employs an asymmetric convolutional encoder–decoder architecture. In the convolutional encoder, two KAN linear layers are integrated to form a spectral KAN block to extract effective shallow nonlinear features from high-dimensional complex HSIs combined with a 2-D CNN layer. In the convolutional decoder, a spatial KAN block is developed combined with a point-wise CNN layer to exploit both spatial and spectral information for image reconstruction. At the same time, to stimulate background recovery and mitigate anomaly expression, an efficient weight initialization mechanism is designed to guide the decoder generation. The main contributions are summarized as follows.

1) To exploit the complex nonlinear features in HSI, inspired by KAN, we devise a `KACNet` by integrating KAN and CNN into an asymmetric convolutional encoder–decoder architecture for HAD tasks. As far as we know, this is the first time to explore the performance of KAN in HAD.
2) To deploy the `KACNet`, a spectral KAN block and a spatial KAN block are designed to be coupled into the convolutional encoder and decoder, respectively. To this end, the spectral KAN block can quickly excavate deep nonlinear features of high-dimensional input to compensate for the limitations of 2-D CNN. The spatial KAN module can efficiently mine spatial nonlinear features in the low-dimensional hidden layer to augment point-wise convolution.
3) To encourage the background recovery in the decoder phase, a weight initialization mechanism is proposed by using hierarchical density-based spatial clustering of applications with noise (HDBSCAN), which can adaptively produce stable background points to initialize the weight of the decoder layer.
4) `KACNet` achieves an overall improvement in detection accuracy and background suppression of 0.69% and 33.58%, respectively, across six real datasets compared to the optimal method in all competitors. These results demonstrate the powerful representation capabilities of

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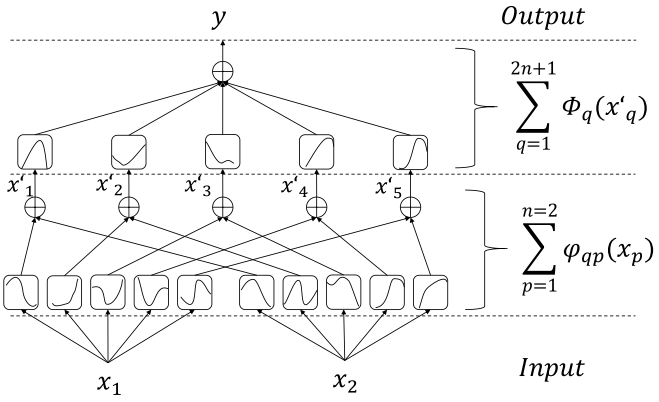


Fig. 1. Diagram of the KAT theorem with dual-variable input. In the first layer, five continuous single-variable functions are adopted for each input variable x_p ($p = 1, 2$) and sum the corresponding output to obtain x'_q ($q = 1, 2, 3, 4, 5$). In the second layer, the other five continuous single-variable functions are applied to x'_q and sum all of them to get the output.

the proposed KACNet, as well as its promising application potential.

The structure of the article is as follows. Section II briefly reviews preparatory knowledge related to the proposed KACNet. Section III presents a comprehensive overview of the newly developed KACNet model. Section IV outlines the experimental configuration and offers a thorough analysis and discussion of the anomaly detection results. Finally, the conclusions are summarized in Section V.

II. Related Work

A. Kolmogorov–Arnold Network

The research of Vladimir Arnold and Andrey Kolmogorov forms the foundation of modern KAN theory. They demonstrated that a multivariate continuous function with a defined boundary can be expressed as a finite composition of continuous single-variable functions, combined with an additive binary operation. Typically, the multivariate function $f(X) = f(x_1, \ldots, x_n)$ can be formulated as follows:

$$f(X) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^{n} \phi_{qp}(x_p) \right) \tag{1}$$

where Φ_q and ϕ_{qp} are single-variable nonlinear functions, n represents the number of input elements, and x denotes the input variable. Based on this definition, Fig. 1 illustrates the structure of this Kolmogorov–Arnold theorem (KAT) with two input variables ($n = 2$). However, these 1-D functions may have nonsmooth characteristics, making them difficult to learn in practice [52]. To address this problem, Liu et al. [53] developed the Kolmogorov–Arnold networks (KANs) by lifting restrictions on network width and depths. Specifically, they defined the KAN layer as a matrix of 1-D functions

$$\Phi = \{\phi_{qp}\} \quad p = 1, 2, \ldots, n, \quad q = 1, 2, \ldots, m \tag{2}$$

where n and m represent the input and output dimensions, respectively. ϕ_{qp} denotes a learnable nonlinear activation function. Then, the more profound KAN can be achieved by stacking multiple KAN layers

$$\mathrm{KAN}(x) = (\Phi_{l-1} \circ \Phi_{l-2} \circ, \ldots, \circ \Phi_1 \circ \Phi_0)x \tag{3}$$

where l is the number of layers. x represents the input features. More details can be accessed at [53].

In contrast to MLP, KAN constructs the network by replacing linear weights with nonlinear functions, referred to as learnable activation functions. This approach improves both the accuracy and interpretability of the KAN-based network, while also enabling it to achieve better performance with fewer parameters in a range of studies [54]. In view of this, KAN-based models have been successfully applied to various domains, such as image processing [55], data analysis [56], and time series processing [57]. In particular, the KAN network demonstrates notable advantages in handling high-dimensional data, as evidenced by its application in HSIs. Jamali et al. [58] developed a hybrid architecture that integrates 1-D, 2-D, and 3-D KANs to achieve precise classification on complex HSIs. Wang et al. [59] utilized the multivariate continuous activation function within KAN to extract hyperspectral features and proposed a KAN-based model named SpectralKAN for change detection in HSIs. Additionally, significant advancements have been achieved in integrating KAN with existing deep-learning networks. For instance, Bresson et al. [60] combined KAN with graph neural networks to perform better on graph regression tasks, while Bodner et al. [61] and Igali and Shamoi [62] integrated KAN with CNN layer and showcased its improvement across various standard datasets.

B. Hierarchical Density-Based Spatial Clustering of Applications With Noise

As a density-based hierarchical clustering algorithm derived from DBSCAN, HDBSCAN [63] integrates hierarchical clustering with advanced techniques for identifying clusters of varying densities. This enables the efficient and automatic grouping of data samples of different sizes and shapes [64]. By computing the mutual reachability distances between all data points, HDBSCAN builds a minimum spanning tree (MST) from these distances. The mutual reachable distance between two points (x_i and x_j) can be calculated as follows:

$$\mathbf{D}_{\mathrm{mreach}}(x_i, x_j) = \max\{\mathrm{Core}(x_i), \mathrm{Core}(x_j), d(x_i, x_j)\} \tag{4}$$

where Core represents the core distance, which can be defined as

$$\mathrm{Core}(x_i) = d(x_i, N^k(x_i)) \tag{5}$$

where d represents the Euclidean distance, while k signifies the minimum number of neighboring sample points of x_i. $N^k(x_i)$ is the kth neighboring point. Then, the density-based cluster hierarchy is created by removing edges from the MST to partition clusters at different density levels. Ultimately, the final clusters are generated by calculating persistence to select stable clusters from the hierarchy and marking other points as noise. By using the concepts of core distance, mutual reachability distance, and hierarchical clustering, HDBSCAN can

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provide better cluster detection for varying densities and automatically handle noise and outliers [65]. Since hyperspectral data are characterized by high dimensionality, complexity, and noise, HDBSCAN's density adaptability and noise processing capability have become an ideal choice for processing such data.

III. Methodology

In this section, the details of the proposed KACNet for HAD are unfolded from the following aspects: overview framework of KACNet for HAD, deployment of KACNet, weight initialization mechanism, and loss function.

A. Overview Framework of KACNet for HAD

The overall flowchart of the proposed KACNet for HAD is presented in Fig. 2, which is composed of three internally interdependent components, i.e., convolutional encoder, convolutional decoder, and the background samples searching for weight initialization. Among them, the background samples searching module is directly related to the structure of the decoder and will decide the output dimension of the encoder as well as the weight initialization of the reconstructive layer of the decoder.

Let the HSI be $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, where C represents the channel dimension, and W and H denote the width and height of HSI, respectively. The core idea of KACNet is to map the original input data into a low-dimensional feature space which is expected to be formed by background features by using a convolutional encoder, and then recover the background by using a relatively lightweight convolutional decoder. In the convolutional encoder, three progressive convolutional encoding modules are designed to extract implicit features from the input HSI $\mathbf{X}$, where the first two modules reduce the number of features by half at a time, and the number of features in the last layer is allocated by the background sample search module. As a result, the model is adaptive and flexible for different datasets. The encoded features are $\mathbf{F}_e^1 \in \mathbb{R}^{H\times W\times (C/2)}$, $\mathbf{F}_e^2 \in \mathbb{R}^{H\times W\times (C/4)}$, and $\mathbf{F}_e^3 \in \mathbb{R}^{H\times W\times M}$, respectively, where M is the number of background samples. In the convolutional decoder, an asymmetric decoder module is designed to directly recover the HSI $\hat{\mathbf{X}}$ from the hidden feature $\mathbf{F}_e^3$. Distinctively, all the modules both in the encoder and decoder are implemented by coupled with the CNN and KAN, which will be described in detail in Section III-B. Because anomalies are more prone to large errors during feature squeeze and expansion and the background initialization mechanism stimulates the reconstruction of the background, the residual map between the predicted $\hat{\mathbf{X}}$ and the input image $\mathbf{X}$ is regarded as the anomaly map of KACNet reasonably. In the implementation of KACNet, the training stage and prediction stage are as follows.

1) Training Stage: Before the training of KACNet, the background samples searching module is adopted first to obtain the representative background samples $\mathbf{W}_o \in \mathbb{R}^{C\times M}$ from $\mathbf{X}$, the detailed process is unfolded in Section III-C. After that, the structure of KACNet can be built and the weight of the point-wise CNN layer in the decoder can be initialized. In the training, the input data $\mathbf{X}$ is input into the encoder to obtain the implicit features, then the decoder recovers the background $\hat{\mathbf{X}}$ from $\mathbf{F}_e^3$. During this process, Adam optimizer will be adopted to dynamically adjust the learning rate and the weight decay, and parameters of the network are continuously optimized by backpropagation to minimize the loss function until the required threshold is met or the epoch reaches the maximum value. To this end, all the parameters of KACNet can be recorded for the prediction.

2) Prediction Stage: In the prediction, the HSI to be tested $\mathbf{X}$ is fed into the trained model with the recorded parameters, which processes the input data based on the feature mapping relationship learned during training to obtain the final predicted $\hat{\mathbf{X}}$. Then, $\mathbf{X}$ and $\hat{\mathbf{X}}$ can be adjusted to 2-D format with the size $C \times N$, where $N = H \times W$ is the number of the pixels in HSI, and the reconstruction residuals can be computed as

$$\mathbf{E} = \left\| \left(\mathbf{X} - \hat{\mathbf{X}}\right)_{:,i} \right\|_2 = \sqrt{\sum_j \left(\mathbf{X} - \hat{\mathbf{X}}\right)_{j,i}^2} \tag{6}$$

where $\| \bullet \|_2$ is the l_2 norm of matrix, while j and i are the jth channel and the ith pixel, respectively.

B. Deployment of KACNet

As presented previously, KAN offers theoretical advantages in expression, generalization, and interpretability over MLP and CNN, making it appropriate for dealing with highly nonlinear and complex problems. Therefore, in the proposed KACNet, KAN is exploited to combine with the convolutional AE model to enhance the interpretability of the net. By substituting the linear weight matrix with a nonlinear activation function, KAN can facilitate faster convergence and improve the model's generalization capabilities. The detailed implementation of the proposed KACNet is illustrated in Fig. 3. The focus is on how to delicately embed the KAN module to achieve a complementary effect with CNN. In the convolutional AE net (CAENet) for HAD shown in Fig. 3(a), the function of the encoder is mainly to extract spatial (the first two layers) and spectral (the third layer) information to reduce the channels from C to M, and then the decoder directly recovers the high-dimensional data from the low-dimensional hidden features. Notably, the weight initialization mechanism allows the decoding layer to consist of a point-wise convolution layer. Based on this network architecture, our natural idea is to add a KAN layer directly behind the CNN layer to further approximate the deep nonlinear features of high-dimensional data. Specifically, we develop a spectral KAN block and a spatial KAN block to support the encoder and decoder as presented in Fig. 3(b), respectively. Thus, KACNet not only enables complementary feature integration but also minimizes the impact of KAN on increased runtime.

1) Spectral KAN Block: The main structure in KAN is the KAN linear layer. Consider the 2-D format of HSI data as $\mathbf{X} \in \mathbb{R}^{N\times C}$. Then, the computation of a KAN linear layer Φ

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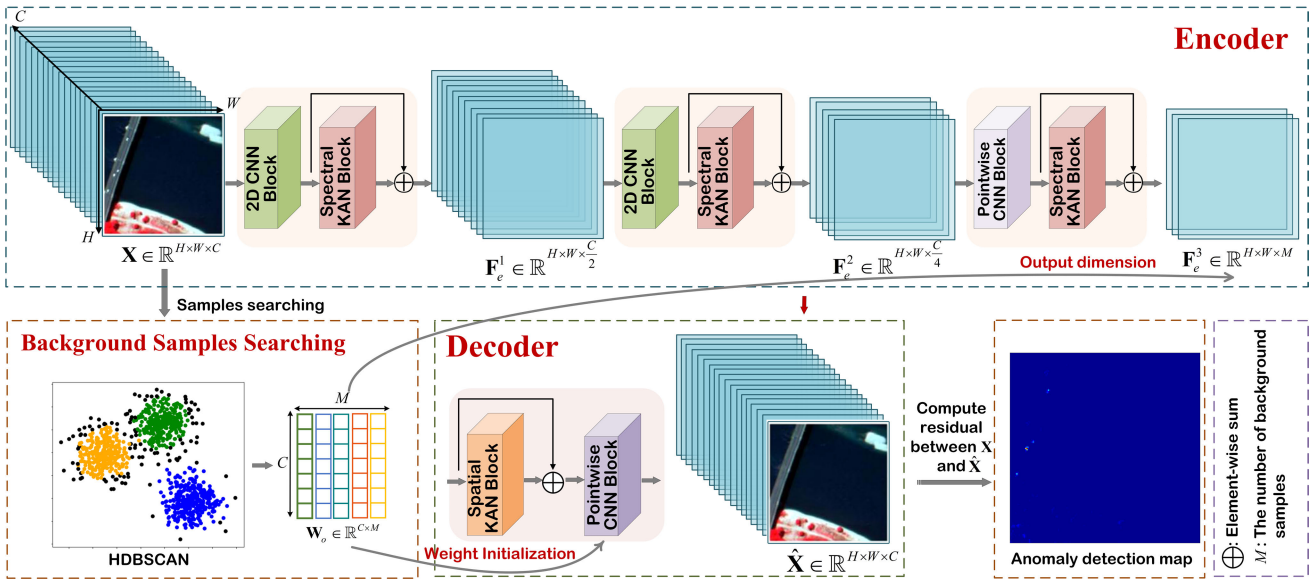


Fig. 2. Flowchart of the proposed KACNet for HAD.

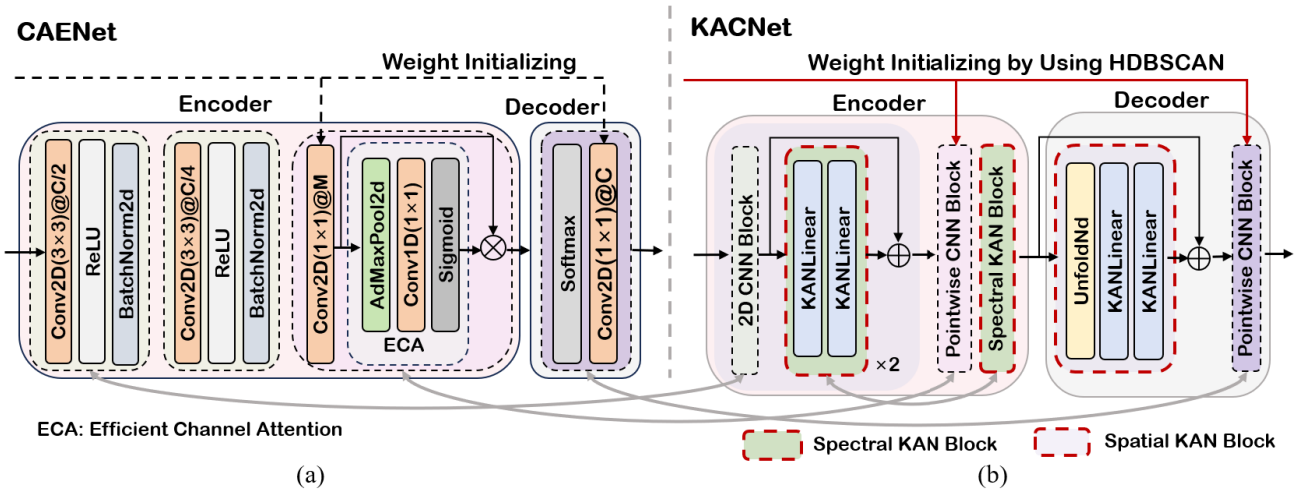


Fig. 3. Detailed architecture of the proposed KACNet. (a) Structure of the convolutional AE net is provided as a baseline model of our proposed KACNet. (b) Main improvement in our KACNet.

can be formulated as follows:

$$\mathbf{X}_{\text{out}} = \underbrace{\begin{pmatrix} \phi_{l11}(\cdot) & \phi_{l21}(\cdot) & \cdots & \phi_{l r_{\text{in}}1}(\cdot) \\ \phi_{l12}(\cdot) & \phi_{l22}(\cdot) & \cdots & \phi_{l r_{\text{in}}2}(\cdot) \\ \vdots & \vdots & & \vdots \\ \phi_{l1 r_{\text{out}}}(\cdot) & \phi_{l2 r_{\text{out}}}(\cdot) & \cdots & \phi_{l r_{\text{in}} r_{\text{out}}}(\cdot) \end{pmatrix}}_{\Phi_l} \mathbf{X}_{\text{in}} \tag{7}$$

where Φ_l represents the lth KAN layer. r_{in} represents the input dimension of lth KAN layer in the input data $\mathbf{X}_{\text{in}}$, and for the first KAN layer, it is equivalent to C. The corresponding output dimension is represented by r_{out}. In this context, ϕ denotes the activation function, which is a nonlinear function applied to the edges (connections) in the network. The spline function is highly effective when representing nonlinear relationships. To ensure it is learnable, a basic function $b(x)$ is introduced as a redundant connection, combining with several learnable parameters to form $\phi(x)$ [53]

$$\phi(x) = w_b b(x) + w_s \text{spline}(x) \tag{8}$$

where $x \in \mathbf{X}$. w_s and w_b are learnable parameters used to better control the shape of the spline function. $b(x)$ typically employs the sigmoid linear unit (SiLU) [66] function

$$b(x) = \text{SiLU}(x) = x * \sigma(x) \tag{9a}$$

$$\sigma(x) = \frac{1}{1+e^{-x}} \tag{9b}$$

where $\sigma(x)$ denotes the logistic sigmoid.

Additionally, spline(x) can be defined as the linear combination of the B-splines, denoted as

$$\text{spline}(x) = \sum_i c_i B_i(x) \tag{10}$$

where the variable c represents an additional learnable parameter, which is refined through the learning process.

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Intuitively, the KAN linear layer is similar to a linear layer of MLP, mainly working on the spectral dimension. The difference is that KAN utilizes the summation of multiple 1-D functions to approximate a multidimensional function, learning deeper nonlinear features under limited network depth than MLP. In this case, we adopt two KAN linear layers to form a spectral KAN block, which is designed as a residual block to compensate for the 2-D CNN layer. Assume the 2-D output features through the 2-D CNN block in the first two layers of the encoder are $\mathbf{F}_{e,0}^{1} \in \mathbb{R}^{N\times(C/2)}$ and $\mathbf{F}_{e,0}^{2} \in \mathbb{R}^{N\times(C/4)}$, respectively, the 2-D output feature through point-wise CNN block in the last layer of the encoder is $\mathbf{F}_{e,0}^{3} \in \mathbb{R}^{N\times M}$. Then, the residual learning process in the encoder can be formulated as

$$\mathbf{F}_{e}^{i'} = \mathbf{F}_{e,0}^{i} + \Phi_{i,2}\left(\Phi_{i,1}\left(\mathbf{F}_{e,0}^{i}\right)\right) \quad i = 1, 2, 3 \tag{11}$$

where $\mathbf{F}_{e}^{i'}$ is the 2-D format of $\mathbf{F}_{e}^{i}$. Notably, the encoder incorporates the efficient channel attention (ECA) layer [67] to enhance the model's ability to focus on the most relevant features by dynamically weighting different channels. This advanced mechanism improves the overall performance of the encoder by ensuring that it can more effectively prioritize important information while processing the input features.

2) Spatial KAN Block: We adopt convolutional KAN to achieve the spatial KAN block. To accomplish this, the UnfoldNd function is first utilized to extract local information and unfold it to a 2-D matrix, where the channel dimension is expanded to include the local information, that is the number of original channels multiply the number of pixels in the kernel. Then, like the spectral KAN block, two KAN linear layers are used to obtain the spatial features. Compared to CNN's weight-based convolution kernels, the difference is the convolution kernel in KAN convolution comprises multiple nonlinear activation functions $\phi(x)$ [defined in (8)]. For the ease of understanding, we use the unfolded form to describe the convolution operation of KAN, a 3×3 KAN convolutional kernel can be defined as follows:

$$\mathbf{K} = \begin{pmatrix} \phi_{11}(\cdot) & \phi_{21}(\cdot) & \phi_{31}(\cdot) \\ \phi_{12}(\cdot) & \phi_{22}(\cdot) & \phi_{32}(\cdot) \\ \phi_{13}(\cdot) & \phi_{23}(\cdot) & \phi_{33}(\cdot) \end{pmatrix}. \tag{12}$$

Then, considering a single-channel image consisting of $n\times n$ pixels, the process of applying KAN convolution is as follows:

$$\underbrace{\begin{pmatrix} y_{11} & y_{21} & \cdots & y_{m1} \\ y_{12} & y_{22} & \cdots & y_{m2} \\ \vdots & \vdots & & \vdots \\ y_{1m} & y_{2m} & \cdots & y_{\mathrm{mm}} \end{pmatrix}}_{\mathbf{X}_{\mathrm{out}}} = \underbrace{\begin{pmatrix} x_{11} & x_{21} & \cdots & x_{n1} \\ x_{12} & x_{22} & \cdots & x_{n2} \\ \vdots & \vdots & & \vdots \\ x_{1n} & x_{2n} & \cdots & x_{nn} \end{pmatrix}}_{\mathbf{X}_{\mathrm{in}}} \mathbf{K} \tag{13}$$

where $\mathbf{X}_{\mathrm{in}} \in \mathbb{R}^{c\times W\times H}$, $W = H = n$ and c represents a specific channel. The variable m represents the dimensionality of the output data. In this case, the padding is set to 0, and the stride size is 1, then $m = n - 2$. Typically, each element in the output data is generated through a convolution operation, where the 3×3 kernel is applied to a window centered on the corresponding element in the input data. For instance, the first element of the output data is computed as follows:

$$y_{11} = \sum_{i=1}^{k=3}\sum_{j=1}^{k=3} \phi_{\mathrm{ij}}\left(x_{\mathrm{ij}}\right). \tag{14}$$

Ultimately, the convolution of the multichannel input feature $\mathbf{X}$ can be accomplished by utilizing a set of KAN convolution kernels.

C. Weight Initialization Mechanism

Albeit AE-based HAD networks primarily exploit feature squeezing and expansion to achieve background recovery and disregard anomalies, the results are not always satisfactory. Theoretically, when the loss function based on the reconstruction error is employed to guide the training of the network, the reconstruction results will get closer and closer to the original input. In this case, the network will pay more and more attention to the parts of the image that are difficult to recover to minimize loss as the number of training times increases. In other words, the network will gradually focus on the reconstruction of the anomalous part, possibly causing the failure of the HAD task. To alleviate this problem, our previous work adopted background-prior to boost the recovery of the background by using the K-means [68], which is a preliminary attempt and presents an impressive performance for accelerating convergence. However, in terms of high-dimensional complex data clustering, K-means is not an optimal choice, especially for HAD tasks. K-means not only requires a predetermined number of clusters and assumes uniform cluster size, but is also sensitive to noise and outliers, leading to unreliable clustering results for HAD. Therefore, in this work, we adopt HDBSCAN to process HSI data, which can effectively recognize complex nonlinear structures and high-dimensional features through local density estimation. More importantly, it can automatically remove noise and outliers at the same time, which is highly matched with the goal of background sample searching.

In detail, we mainly use three settings to select robust background samples. The first is to control the *size of the minimum cluster*, if the number of points in the cluster is less than 0.1% of all points, the cluster will be defined as an ineffective class, which can prevent the generation of anomalous clusters to some extent. The second is to control the *quality of the core points*, if more than 50 points are near a core point, this core point will be regarded as high-quality, which simultaneously excludes noise points and anomalous points. The third is the *cluster selection pattern*, where we use leaf nodes of hierarchical trees as the clustering result, which produces finer background classes. Additionally, the HDBSCAN only assigns category labels to the input data without clustering centers, that is $\mathbf{X}_{\mathrm{class}} \in \mathbb{R}^{1\times N}$, which includes one invalid class with the value of -1 and M valid classes with the value $\{0, 1, \ldots, M-1\}$. Next, we traverse the M valid classes to get the final background samples as follows:

$$\mathbf{W}_o[:, i] = \frac{1}{m_i}\sum_{j=1}^{m_i}(\mathbf{X}_{2D}[:, \mathbf{X}_{\mathrm{class}} == i-1]) \tag{15}$$

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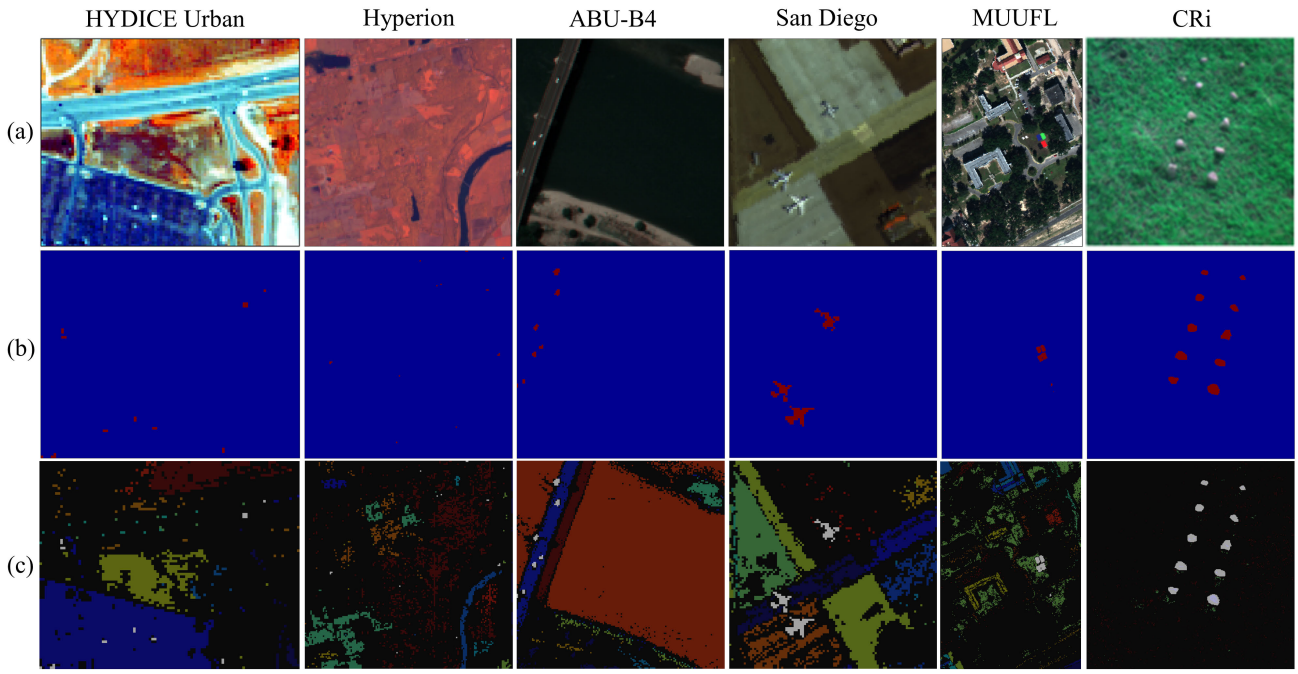


Fig. 4. (a) Pseudocolor HSI scene, (b) anomaly reference map, and (c) corresponding clustering results for the selected six datasets. Concretely, (c) shows the overlay display of the anomaly reference map and clustering results, where the black region is the invalid class and the gray region is the anomaly after the overlay display. If the anomaly region overlays the valid classes, other colors are displayed, otherwise it is pure gray.

where m_i is the number of pixels in the ith cluster and $\mathbf{X}_{2D}$ is the 2-D format of the input HSI data. When computing (15) for all the clusters, the $\mathbf{W}_o$ can be obtained and assigned to the weight of the point-wise CNN layer in the decoder.

D. Loss Function

In AE-based models, the mean absolute error (MAE) is a common metric for evaluating reconstruction loss. Its limited sensitivity to outliers enables the AE to concentrate more effectively on reconstructing normal data. Accordingly, the loss computed during `KACNet` training can be defined as

$$\mathcal{L}\text{oss} = \text{MAE}(\mathbf{X}, \hat{\mathbf{X}}) = \frac{1}{\text{CWH}} \sum_{c}^{C} \sum_{i}^{W} \sum_{j}^{H} \left| x_{cij} - \hat{x}_{cij} \right| \quad (16)$$

where $\mathbf{X}$ and $\hat{\mathbf{X}}$ denote the input and reconstructed HSIs.

IV. Experiments

This section evaluates and analyzes the performance of the proposed `KACNet` by comparing it with nine existing classic HAD methods, utilizing six real-world hyperspectral datasets. Additionally, the effectiveness of the employed KAN, as well as the weight initialization mechanism, is further validated through ablation studies.

A. Datasets

The dataset utilized comprises six real hyperspectral scenes, each presenting distinct challenges. These scenes differ in image dimensions, spectral bands, as well as the complexity and characteristics of both the background and the objects within them. The pseudocolor, anomaly reference maps, and clustering results of HDBSCAN for all datasets are presented in Fig. 4.

1) HYDICE Urban [69]: This dataset comprises of an image captured in urban environments during the hyperspectral digital image collection experiment (HYDICE) in 1995. It features a spatial resolution of 1.56 m and a spectral resolution of 10 nm. Certain rooftops and vehicles have been identified as anomalous targets. This image comprises 175 spectral bands, each with a resolution of 80 × 100 pixels, with anomalous targets occupying 21 pixels.

2) Hyperion: This dataset pertains to a representative agricultural region, collected in Indiana, USA, using Hyperion imaging sensors in 2008. Small objects, including silos and rooftops that merge with the background, are defined as anomalies. The experimental data comprise 155 spectral bands with 150 × 150 pixels, featuring a spectral resolution of 10 nm and a spatial resolution of 10 m.

3) ABU-B4 [70]: This data identify vehicles situated on bridges as anomaly targets. The background is diverse and complex, comprising elements such as rivers, bridges, vegetation, and exposed terrain. The data used in the experiment is sourced from the Airport-Beach-Urban (ABU) dataset, consisting of the image sized at 150 × 150 pixels with 102 spectral bands. This image was captured in Pavia's central region using the reflective optics system imaging spectrometer (ROSIS-03) sensor, which offers a spatial resolution of 1.3 m.

4) San Diego: This dataset features a classic airport scene, captured by the AVIRIS sensor at San Diego Airport, California, USA. The original image measures 400 × 400 pixels and contains 189 effective spectral bands, with a spatial resolution of 3.5 m. In our experiments, a 100 × 100 pixel sub-image was extracted, highlighting three aircraft as anomalous objects.

5) MUUFL [71]: This dataset was collected using a Compact Airborne Spectrographic Imager (CASI-1500) with a

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TABLE I
ABLATION STUDIES ON THE ARCHITECTURE OF KACNET IN AUC. VALUES IN BOLD INDICATE THE BEST RESULTS

	Spectral KAN	Spatial KAN	HYDICE Urban	Hyperion	ABU-B4	San Diego	MUUFL	CRi
KACNet-Baseline	✗	✗	0.9962	0.9994	0.9854	0.9891	0.9985	0.9906
KACNet-Spectral	✓	✗	0.9962	0.9995	0.9853	0.9902	0.9989	0.9908
KACNet-Spatial	✗	✓	0.9965	0.9993	0.9886	0.9879	0.9982	0.9906
KACNet	✓	✓	**0.9974**	**0.9996**	**0.9931**	**0.9915**	**0.999**	**0.9913**

TABLE II
ABLATION STUDIES OF KACNET INVOLVING GUIDANCE WITH K-MEANS, HDBSCAN, AND UNDER THE ABSENCE OF ANY GUIDANCE IN AUC. VALUES IN BOLD INDICATE THE BEST RESULTS

	Kmeans	HDBSCAN	HYDICE Urban	Hyperion	ABU-B4	San Diego	MUUFL	CRi
KACNet-NoGuided	✗	✗	0.9922	0.9946	0.9687	0.9914	0.9944	**0.992**
KACNet-K-means	✓	✗	0.9953	0.9961	0.9843	**0.9916**	0.9969	0.9755
KACNet-HDBSCAN	✗	✓	**0.9974**	**0.9996**	**0.9931**	0.9915	**0.999**	0.9913

spectral resolution of 10 nm. The anomalous targets in the scene are some artificial fabric panels. The original data were captured in November 2010 at the campus of the University of Southern Mississippi-Gulfport (MUUFL), featuring a resolution of 325 × 220 pixels and comprising 64 spectral bands, including 279 pixels identified as anomalous.

6) CRi [72]: This dataset identifies ten large rocks in a grassy scene as anomaly targets. The image was captured using a Nuance Cri sensor with a spectral resolution of 10 nm. The image consists of 400 × 400 pixels and includes 46 bands, with multiple anomalous targets covering 2216 pixels. From Fig. 4(c), HDBSCAN effectively excludes anomalies from valid classes, except in the case of CRi. This is primarily due to the large scale of the scene and the high number of anomalies which leads to anomalies being identified as a valid class.

B. Experimental Setup

1) Methods and Metrics for Evaluation: We conducted a comparison of the proposed KACNet method against nine existing approaches, namely: 1) *RX* [12]; 2) *CRD* [21]; 3) *LRASR* [26]; 4) *PTA* [29]; 5) *Auto-AD* [38]; 6) *RGAE* [33]; 7) *GAED* [34]; 8) *PDBSNet* [40]; and 9) *GT-HAD* [49]. Meanwhile, to comprehensively evaluate the detection performance, an in-depth investigation is conducted by using five classic metrics. These include the receiver operating characteristic (ROC) curve [73], which offers a visual depiction of the balance between the true-positive rate and the false-positive rate, and the area under the ROC curve (AUC), a concise numerical summary that delivers an aggregated measure of performance over all possible thresholds. Additionally, the *box plot* [74] is deployed, which is a visual tool that summarizes the distribution, variability, and outliers within the results. The analysis is further refined by using the square error ratio (SER) to measure the difference between predicted and actual values, and the area error ratio (AER) to assess cumulative spatial errors, offering insights into model precision and accuracy [75], [76].

2) Implementation Details: The experiments were conducted using MATLAB 2016b for traditional methods, while deep learning-based methods employed a cluster of NVIDIA GeForce RTX 3070 GPUs and were performed under the PyTorch framework. In the proposed KACNet, the KAN component is realized through the application of EfficientKAN.[1] During the training phase, KACNet employs the Adam optimizer for gradient optimization, with a learning rate of $1e^{-2}$ and a decay rate of $1e^{-8}$. After 500 iterations, the model parameters are stored for testing purposes.

C. Ablation Study

This article accomplishes precise HAD through two aspects. First, by integrating KAN with CNN to form an AE network, the model's interpretability and detection accuracy are improved. Additionally, the HDBSCAN is employed to efficiently extract background information, serving as prior knowledge to guide the model in the training process. To ascertain their efficacy, ablation studies are conducted independently on various datasets.

1) Effectiveness of Integrating CNN and KAN: In the proposed KACNet, the spectral KAN block is utilized within the encoder, while the spatial KAN block is employed in the decoder. Table I reports the AUC values for the given ablation configurations for the model architecture. Among them, KACNet-baseline represents the baseline structure, which utilizes only CNN to conduct anomaly detection. Building upon KACNet-Baseline, KACNet-spectral incorporates the spectral KAN block into the encoder. Similarly, KACNet-spatial is derived from KACNet-baseline, but integrates the spatial KAN block exclusively in the decoder. Based on the results, a close examination of the KACNet-spectral and KACNet-spatial architectures reveals that they are capable of making improvements across several datasets. But in certain datasets, such as the KACNet-spectral on dataset ABU_B4 and the KACNet-spatial on dataset San Diego, its performance shows below that of the baseline structure. However, the KACNet demonstrates substantial improvements across all datasets, indicating that the integration of both spectral and spatial blocks can achieve more desirable and stable performance.

[1]Source: https://github.com/Blealtan/efficient-kan

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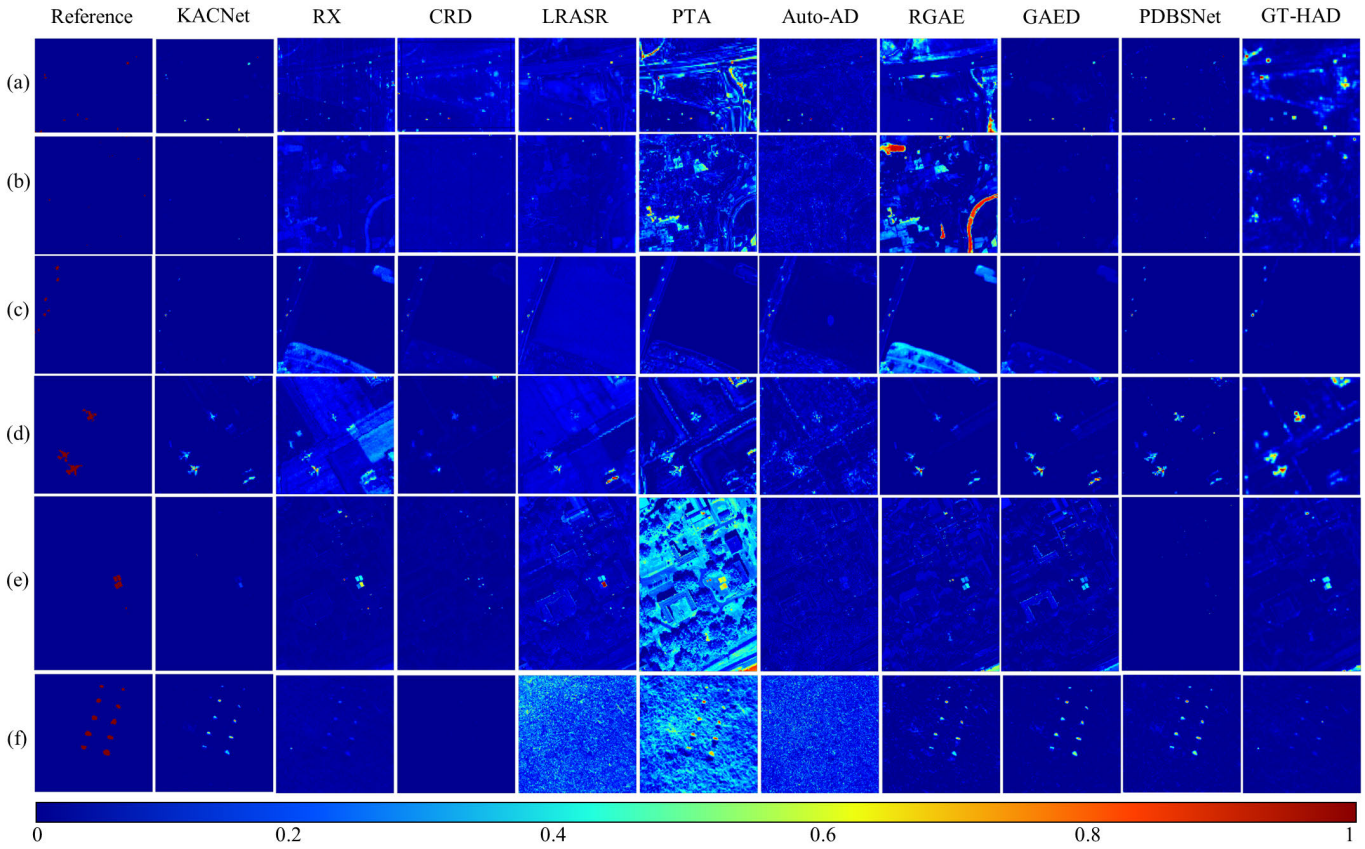


Fig. 5. Anomaly maps of all the considered methods for the selected six datasets. (a) HYDICE Urban, (b) Hyperion, (c) ABU-B4, (d) San Diego, (e) MUUFL, and (f) CRi.

2) Background Prior Guidance: Referring to the prior information of the input data in model training is an effective way to improve performance. The prior information in this study is utilized to initialize both the encoding dimensions of the encoder and the weights of the output layer in the decoder. Table II displays the AUC values for different datasets, employing three separate initialization approaches. The KACNet-NoGuided utilizes fixed encoding dimensions and initializes all decoding layers randomly, which results in improved performance exclusively on the CRi dataset. This is primarily because it is challenging to match a fixed encoding dimension with input data that has features of varying dimensions. Simultaneously, the parameters are initialized based on the characteristics of the input data using K-means or HDBSCAN, which significantly enhances the model's training guidance capabilities and improves detection accuracy. Specifically, in comparison to the K-means, HDBSCAN delivers higher quality clustering results and is better suited to handle the complexity of hyperspectral data, and generally exhibits superior performance across a majority of datasets.

D. Results and Analysis

This section presents the anomaly detection results of the proposed KACNet and nine competitors on selected six HSI datasets. The anomaly maps are displayed in Fig. 5, the boxplots are shown in Fig. 6, and the ROC curves are plotted in Fig. 7, and the quantitative assessment of corresponding AUCs, SER, AER, and time efficiency are listed in Table III.

1) Visualization Performance: From the anomaly maps as shown in Fig. 5, intuitively, the background residuals in the anomaly map of proposed KACNet are almost negligible compared with the reference map, especially in HYDICE Urban, Hyperion, ABU-B4, MUUFL, and CRi datasets, indicating the powerful representation capability of the newly designed network. In the San Diego dataset, some tiny backgrounds fail to be properly reconstructed, and other methods suffer from the same problem. Notably, the GAED and PDBSNet show competitive results on the HYDICE, ABU-B4, San Diego, and CRi datasets, with slightly less background suppression. In addition, although the background errors in the anomaly map of GT-HAD are large, the anomalies are highlighted more than in other methods, which can be observed in HYDICE Urban, Hyperion, ABU-B4, San Diego, and MUUFL datasets. In contrast, Auto-AD, RX, CRD, and LRASR underperform both in background recovery and anomaly highlight. PTA and RGAE, on the other hand, show poor representation ability of the network. In general, the deep learning-based methods can produce better detection maps than the traditional methods.

Moreover, boxplots depicted in Fig. 6 simultaneously verify the above observation, in which the black box is shorter and closer to 0, the better the background is restored and the red box is shorter and closer to 1, the better the anomaly is highlighted. Moreover, the larger the distance between the

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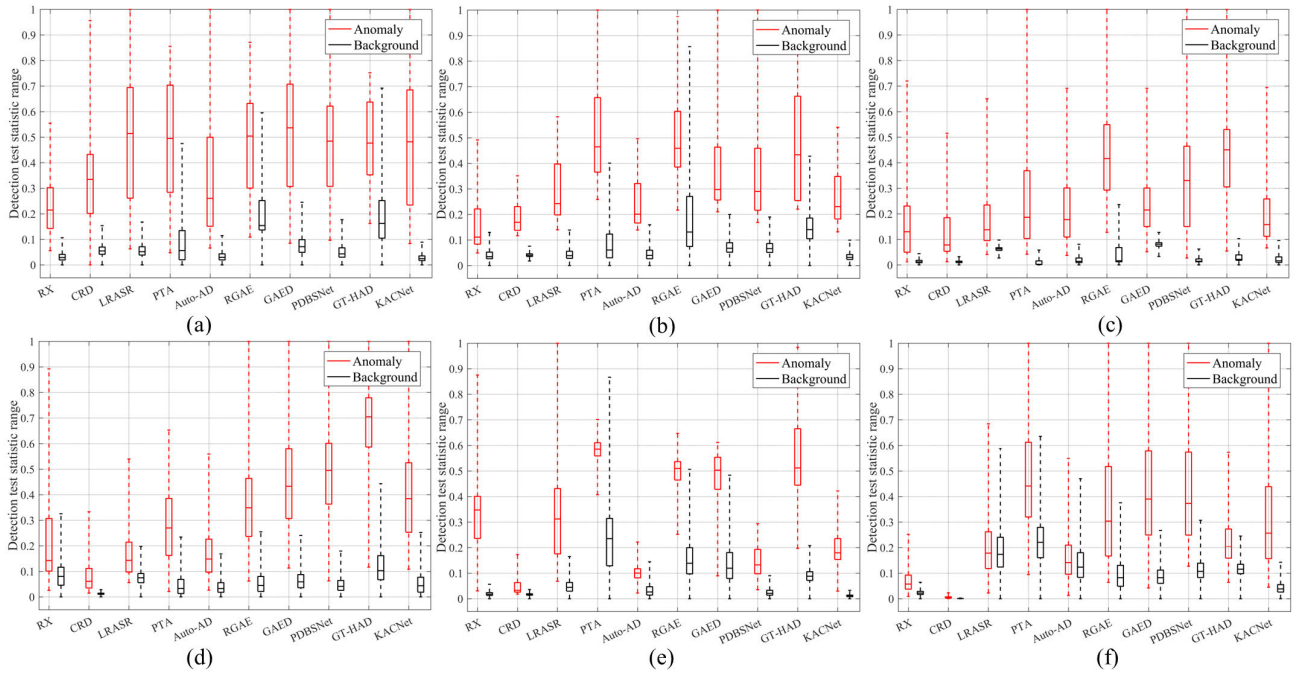


Fig. 6. Boxplots of all the considered methods for the selected six datasets. (a) HYDICE Urban, (b) Hyperion, (c) ABU-B4, (d) San Diego, (e) MUUFL, and (f) CRi.

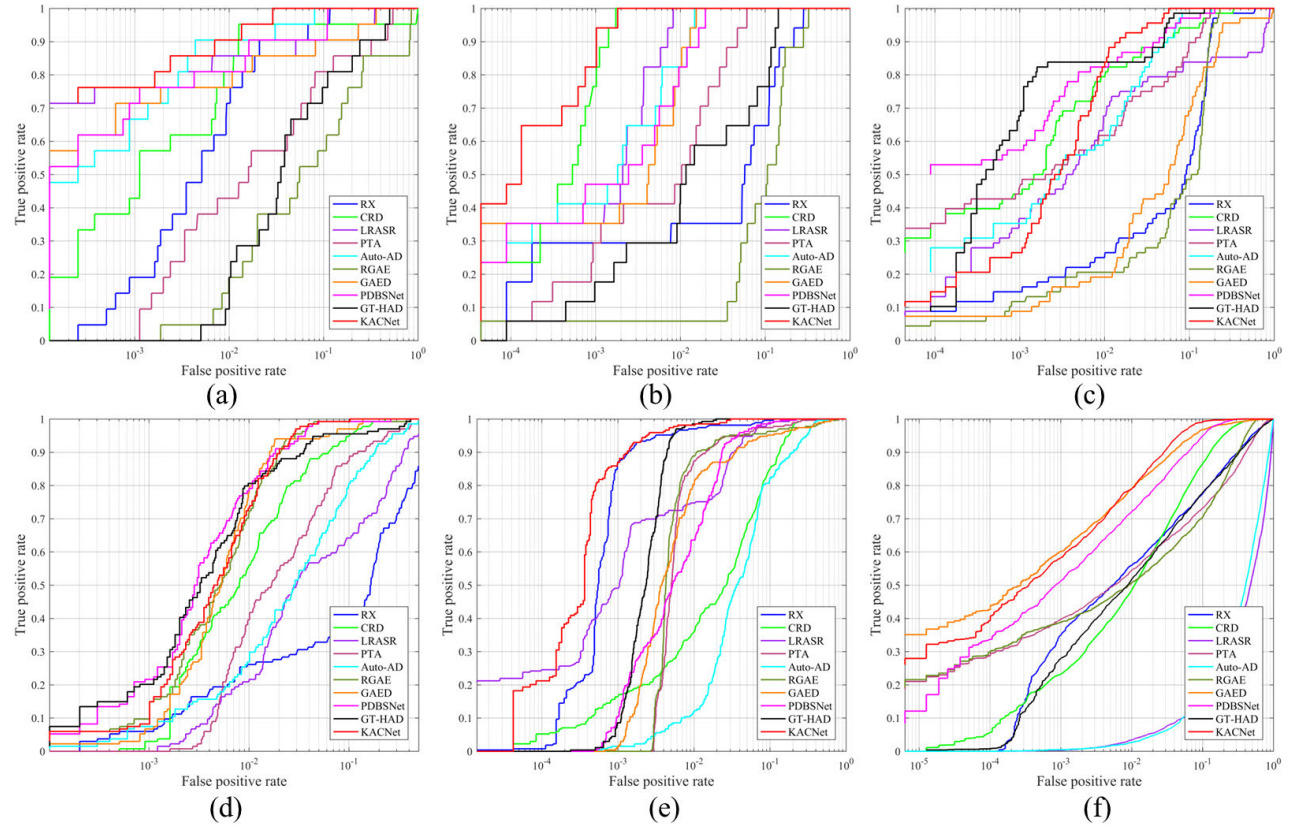


Fig. 7. ROC curves of all the considered methods for the selected six datasets. (a) HYDICE Urban, (b) Hyperion, (c) ABU-B4, (d) San Diego, (e) MUUFL, and (f) CRi.

red box and the black box, the higher the degree of separation between the background and anomalies. In this regard, the proposed KACNet can basically separate background and anomaly for all the selected datasets and perform outstandingly on the HYDICE Urban, Hyperion, and CRi datasets, which can distinguish the anomalies as well as suppress the background.

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TABLE III

AUC, SER, AND AER OF ALL THE CONSIDERED METHODS FOR THE SELECTED SIX DATASETS. VALUES IN BOLD AND UNDERLINED INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Dataset		RX	CRD	LRASR	PTA	Auto-AD	RGAE	GAED	PDBSNet	GT-HAD	KACNet
HYDICE Urban	AUC ↑	0.9857	0.9499	0.9686	0.9122	0.9939	0.8127	0.9663	0.9876	0.9049	**0.9974**
	SER ↓	0.3815	0.5922	0.5300	2.4676	0.2933	6.4371	0.9128	0.5513	5.4652	**0.2253**
	AER ↑	1.2528	1.4489	1.7628	1.7338	1.4699	1.4697	**1.8148**	1.7308	1.5239	1.7997
	Time (s)	0.35	17.62	52.68	49.98	50.51	150.11	68.21	71.98	10.21	69.03
Hyperion	AUC ↑	0.9227	0.9994	0.9975	0.9845	0.9962	0.8789	0.9949	0.995	0.9562	**0.9996**
	SER ↓	0.3600	0.2306	0.3111	1.7780	0.3562	8.0864	0.7697	0.6908	3.1014	**0.2016**
	AER ↑	1.1154	1.2439	1.3753	**1.8495**	1.3041	1.5572	1.4559	1.4065	1.5574	1.3441
	Time (s)	0.41	52.51	138.91	101.75	20.51	380.20	195.83	156.04	54.85	190.25
ABU-B4	AUC ↑	0.9075	0.9819	0.8773	0.9706	0.9832	0.9042	0.8968	0.9884	0.9902	**0.9932**
	SER ↓	0.5990	0.2529	0.6032	0.3172	0.3190	3.8779	1.4705	**0.2362**	0.3112	0.3091
	AER ↑	1.1834	1.1586	1.1619	1.3091	1.2762	1.5773	1.2146	1.4400	**1.6474**	1.2452
	Time (s)	0.41	225.21	431.56	144.74	25.04	501.50	245.78	150.86	29.86	106.63
San Diego	AUC ↑	0.776	0.9786	0.8778	0.9476	0.9316	0.9891	0.9884	0.9912	0.9797	**0.9915**
	SER ↓	2.1496	1.1893	1.7431	1.4721	1.2930	1.1857	1.3595	**1.0312**	3.1118	1.1287
	AER ↑	1.1531	1.0773	1.1268	1.3132	1.1602	1.5091	1.6965	1.8253	**2.6305**	1.5803
	Time (s)	0.27	10.69	53.23	57.75	16.00	190.52	86.48	79.92	14.91	66.29
MUUFL	AUC ↑	0.9977	0.9535	0.9897	0.9872	0.9291	0.9795	0.976	0.9881	0.9973	**0.9991**
	SER ↓	**0.2439**	0.3672	0.5525	6.6169	0.5177	3.0154	2.7198	0.4040	1.2213	0.2633
	AER ↑	1.5040	1.0622	1.6118	1.8270	1.0731	1.6641	1.6281	1.1381	**2.0572**	1.2559
	Time (s)	0.55	57.48	1420.76	96.31	49.12	1034.75	454.12	423.8	655.92	464.59
CRi	AUC ↑	0.9135	0.9571	0.5327	0.8936	0.5698	0.9109	0.9839	0.9821	0.9078	**0.9913**
	SER ↓	1.2713	1.3668	5.3501	5.9347	3.5632	1.9688	1.5122	2.0411	2.3003	**0.9707**
	AER ↑	1.0480	1.0061	1.0159	1.4725	1.0268	1.3994	**1.5712**	1.5265	1.1377	1.3912
	Time (s)	0.73	526.1	1133.9	190.22	106.16	1651.31	923.24	865.93	428.95	673.04
Average	AUC ↑	0.9172	0.9701	0.8739	0.9493	0.9006	0.9126	0.9677	0.9887	0.9560	**0.9954**
	SER ↓	0.8342	0.6665	1.5150	3.0978	1.0571	4.0952	1.4574	0.8258	2.5852	**0.4427**
	AER ↑	1.2095	1.1662	1.3424	1.5842	1.2184	1.5295	1.5635	1.5112	**1.7590**	1.4361
	Time (s)	0.45	148.27	538.51	106.79	44.56	651.40	328.94	291.42	199.12	261.64

As for ABU-B4, San Diego, and MUUFL, GT-HAD generates a better separation degree of background and anomaly than the proposed KACNet, which is mainly because of the higher highlighting effect of GT-HAD. Although KACNet significantly enhances the network's feature extraction and representation capabilities by incorporating KAN, at the same time, it inevitably strengthens the representation of anomalies, making them less effectively highlighted.

2) Anomaly Detection Accuracy and Efficiency: In terms of quantitative evaluation, we analyze the accuracy of all methods in conjunction with ROC curves presented in Fig. 7. Commonly, the closer the ROC curve is to the upper left corner of the axis, the better the detection result is. To this end, the ROC curve of proposed KACNet is obviously superior to other methods for HYDICE Urban, Hyperion, and MUUFL datasets, in which the detection rate is almost higher than other methods with the increase of false alarm rate. In ABU-B4 and CRi datasets, when the false alarm rate is larger than $1e^{-2}$, the detection rate of KACNet quickly grows up and is the first one to reach 1. As for the San Diego dataset, the detection performance of KACNet, GT-HAD, PDBSNet, GAED, and RGAE are equally matched. After that, KACNet, PDBSNet, and RGAE exhibit higher detection rates as the false alarm rate increases. The corresponding AUC in Table III accurately portrays the performance differences between the different methods. Among them, the detection accuracy of KACNet remarkably outperforms the other methods for HYDICE Urban, ABU-B4, MUUFL, and CRi datasets, which are 99.74%, 99.32%, 99.91%, and 99.13%, respectively. The detection accuracy of KACNet slightly outperforms the other methods for Hyperion and San Diego datasets, which are 99.96% and 99.15%, respectively. Overall, the detection accuracy of all the considered methods can be roughly ranked as: KACNet> PDBSNet> CRD> GAED> GT-HAD> PTA> RX> RGAE> Auto-AD> LRASR.

In addition, we also compute the SER and AER scores for all the methods, in which SER can measure the effectiveness of background suppression and anomaly enhancement on pixel-by-pixel, while AER is used to assess the degree of separation between the background and anomalies. The latter is achieved by plotting the detection rate curve and false alarm rate curve under the different thresholds and calculating the ratio of the areas under the two curves. Lower SER and higher AER indicate better detection. As shown in Table III, the proposed KACNet can obtain good scores for all the datasets on SER, whilst the AER score is not satisfied. Through the

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deep analysis, the principal reasons are summarized in two aspects. The first is that the value of SER is largely determined by whether the background pixel reconstruction is good or bad since the number of background pixels dominates in HSI. Therefore, a small portion of the large background residuals will also lead to high SER scores, and lower SER means better background recovery. In contrast, AER relies more on whether the anomaly is highlighted or not, given that in that case, the detection rate rises rapidly. Consequently, SER and AER scores are consistent with the previous analysis in terms of visualization performance, which is why GT-HAD and PTA can obtain good scores on AER.

Nevertheless, the inclusion of KAN layers leads to an increase in training time. Among the compared algorithms, the proposed method exhibits moderate runtime, with computation time scaling proportionally with data size, similar to other methods. However, given the high precision achieved in anomaly detection, this increase is acceptable relative to other approaches.

V. Conclusion

This article mainly explores the performance of KAN in HAD tasks by embedding it into a convolutional AE network, i.e., `KACNet`. Specifically, a spectral KAN block is designed to enhance the feature extraction capabilities of the convolutional encoder, and at the same time, a spatial KAN block is developed in the decoder including a point-wise CNN layer to mine the spatial nonlinear features efficiently. Moreover, a weight initialization mechanism is devised to boost the background recovery by using HDBSCAN which can automatically remove noise and outliers. Extensive experiments on six real datasets demonstrate that the introduction of KAN significantly improves the feature mining and representation capabilities of the network, excelling in background reconstruction. However, this is KAN's initial attempt in HAD and inevitably has flaws, that is how to suppress the powerful representation of KAN on anomalies reconstruction so that the anomalies can be highlighted significantly. In the subsequent work, we will consider adopting gate control into KAN or mask strategy to reconstruct background and anomaly differentially.

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